A Reconfigurable Co-Simulation Framework for Dynamical Analysis of Grid-Connected Wave Energy Converters

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Grid-connected Wave Energy Converters integrate seamlessly into the power grid, maintaining dynamic stability under transient disturbances while providing marginal smoothing and damping benefits.

Motivation

Wave Energy Potential

Ocean waves contain an enormous, yet largely untapped, source of renewable energy. The U.S. technical potential for marine resources exceeds **2,300 TWh per year**, with wave energy dominating most coastal regions *(Figure 1)*. Harnessing even a fraction of this resource could contribute substantially to decarbonizing the power grid while providing generation that complements wind and solar output.

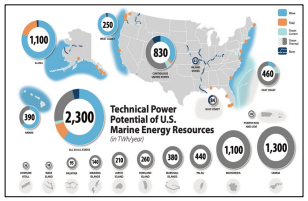


Figure 1. *Technical power potential of U.S. marine energy resources (adapted from Kilcher et al., 2021). Wave energy alone represents over 1,100 TWh/yr of recoverable resource.*

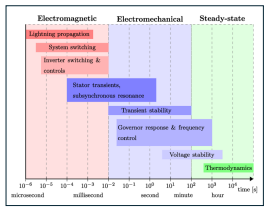


Figure 2. Time-domain scales in power system analysis (Meegahapola et al., 2020). This work focuses on the electromechanical range—seconds to tens of seconds—typical of dynamic stability simulations

Dynamic Power System Modeling

To integrate such emerging renewables, engineers rely on multi-timescale simulation methods that capture how electrical systems respond to disturbances—from microsecond electromagnetic events to minute-scale electromechanical dynamics (Figure 2). Our study operates in the dynamic (electromechanical) time domain, where frequency, voltage, and power flows evolve over seconds after faults or set-point changes—crucial for understanding how inverter-based WECs affect grid stability.

Example

System Overview

To demonstrate the co-simulation workflow, we compared the standard **IEEE-14-bus** test system against a **modified IEEE-15-bus** configuration that includes a dedicated **WEC bus** (Bus 15) connected to **Bus 8** via an additional transmission line.
This representation allows direct evaluation of how wave-energy injection influences grid dynamics during both steady and faulted operation.

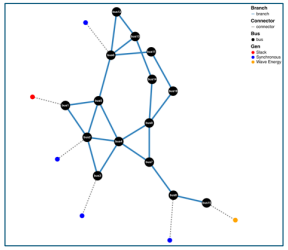


Figure 3. *Network representation of the IEEE-15 system generated with MarinePowerDynamics.jl. Blue nodes denote generators, red and orange nodes indicate loads and the WEC device, respectively.*

WEC-Sim Hydrodynamic Input

Hydrodynamic behavior was modeled in WEC-Sim using the Reference Model 3 (RM3) point-absorber under irregular sea states representative of the Oregon coast (H = 2.5 m, T = 8s).
The simulation outputs time-series data for wave elevation and PTO mechanical power, exported as JSON files for the electrical co-simulation.
These data provide the "raw" mechanical power available from the ocean prior to conversion through the Linear PTO inverter model.

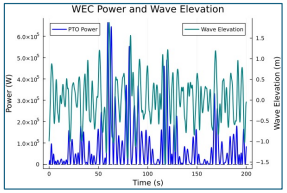
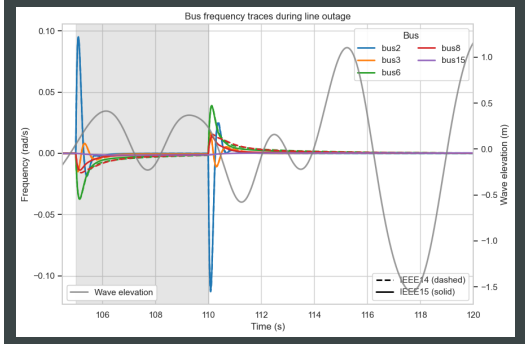


Figure 4. *Time-series of mechanical PTO power (left axis, W) and wave elevation (right axis, m) obtained directly from the* WEC-Sim *simulation.*

Dynamic Simulation Setup

Using MarinePowerDynamics.jl, the Linear PTO model (based on a grid-forming inverter) was coupled with both test systems.
Each run included a 100 s warm-up, a 60 s dynamic window, and a 5 s line outage (Bus 2–5, 105–110 s) to evaluate frequency and voltage recovery.
This setup enables side-by-side comparison of grid response with and without the integrated WEC device.

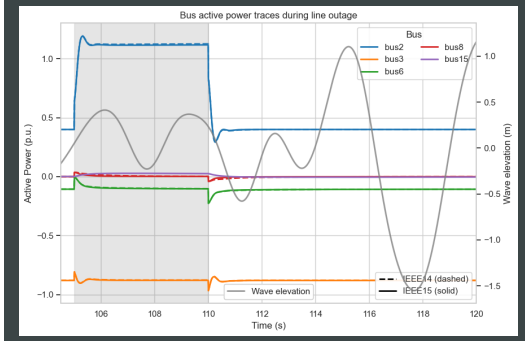
MarinePower Dynamics.jl



Frequency Response

A 5-second fault (Bus 2–5) introduces small frequency deviations across generator buses. Both IEEE-14 (dashed) and IEEE-15 (solid) systems maintain stability, with oscillations remaining within ±0.1 rad/s and damping out within ~2 s. The WEC case mirrors baseline behavior, verifying stable coupling of inverter-based marine generation to the bulk grid.

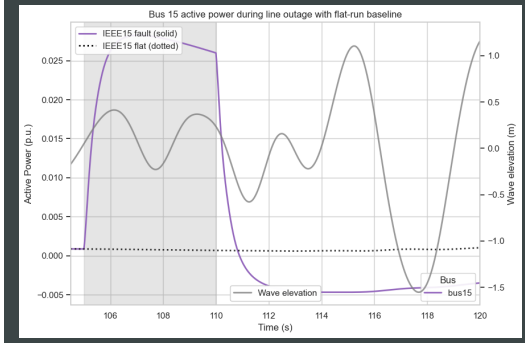
Figure 6. *Bus frequency traces during fault.*



Active power Response

Transient power swings are well contained. The IEEE-15 configuration shows slightly smoother recovery and reduced overshoot at generator buses—indicating minor inertial and damping support from the Linear PTO inverter. These results suggest that WEC integration neither burdens nor destabilizes existing generation.

Figure 7. *Bus active-power traces during fault.*



WEC Bus Response

The WEC injects up to 0.025 p.u. active power under normal operation, following wave elevation patterns. During the 5 s line fault, injection briefly collapses to zero, then recovers immediately after fault clearance, maintaining synchronism with the grid.
This confirms that the WEC model's dynamic response behaves predictably under transient network disturbances.

Figure 8. *WEC (Bus 15) power injection*

The co-simulation results demonstrate that coupling the RM3 WEC through the Linear PTO inverter introduces no instability or adverse interaction with synchronous generation. Both steady-state and faulted conditions show nearly identical dynamic responses between the IEEE-14 (baseline) and IEEE-15 (with WEC) systems. The WEC behaves predictably—temporarily reducing output during a network fault and resuming synchronized operation within seconds of recovery—highlighting its potential to contribute to future inverter-dominated grids without compromising reliability.

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Link to paper

Github

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Link to poster

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Methods

Overview:

The framework couple hydrodynamic simulation in WEC-Sim with dynamic grid simulation in MarinePowerDynamics.jl.

Time-series from WEC-Sim feed into the Linear Power Take-Off (PTO) model, which represents the WEC as a grid-forming inverter based on the VSIMinimal formulation.

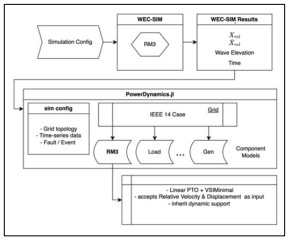


Figure 5. Co-simulation workflow linking hydrodynamic and electrical domains through WEC-Sim, the Linear PTO model, and MarinePowerDynamics.jl.

WEC-Sim Hydrodynamics

WEC-Sim provides the mechanical response of the RM3 device, capturing relative displacement X_{rel} and velocity $\dot{X}_{rel}$

The PTO force is computed as:

$$F_{PTO} = -K_{PTO}X_{rel} - C_{PTO}\dot{X}_{rel}$$

Mechanical power extracted from wave motion:

$$P_{mech} = F_{PTO}\dot{X}_{rel}$$

Linear PTO Conversion

The mechanical power is converted to electrical power through an efficiency factor η and converted to per-unit before injection

Grid-Forming Inverter

The Linear PTO operates as an AC voltage source that regulates **frequency** and **voltage** via droop control:

$$\dot{\omega_a} = \frac{1}{\tau_{P,a}}\left[-\omega_a - K_{P,a}(\mathrm{Re}(u_a i_a^*) - P_a)\right]$$

$$\tau_{Q,a}\dot{v_a} = -v_a + V_{r,a} - K_{Q,a}(\mathrm{Im}(u_a i_a^*) - Q_a)$$

These dynamics allow the inverter to adjust its output based on real and reactive power deviations, maintaining stable operation during disturbances.

Results cont.

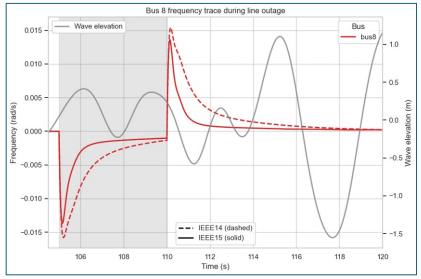


Figure 9. Bus 8 frequency trace during line outage.

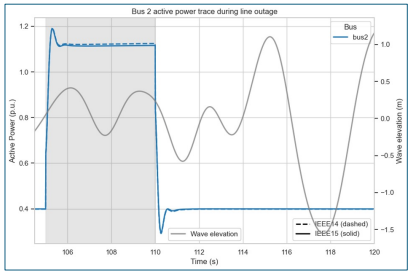


Figure 10. Bus 2 active power trace during line outage.

Interpretation

Together, these localized results confirm that:

- The WEC connection at Bus 8 does not destabilize nearby frequency behavior; it introduces slight damping instead.
- The generator at Bus 2, despite being on the faulted corridor, recovers more smoothly when the WEC is present.
- The overall system response remains well within normal operational limits, verifying that the Linear PTO inverter interacts benignly with grid dynamics.